



Research article

Use of interactive data visualization in multi-objective forest planning

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ABSTRACT

Common to multi-objective forest planning situations is that they all require comparisons, searches and evaluation among decision alternatives. Through these actions, the decision maker can learn from the information presented and thus make well-justified decisions. Interactive data visualization is an evolving approach that supports learning and decision making in multidimensional decision problems and planning processes. Data visualization contributes the formation of mental image data and this process is further boosted by allowing interaction with the data. In this study, we introduce a multi-objective forest planning decision problem framework and the corresponding characteristics of data. We utilize the framework with example planning data to illustrate and evaluate the potential of 14 interactive data visualization techniques to support multi-objective forest planning decisions. Furthermore, broader utilization possibilities of these techniques to incorporate the provisioning of ecosystem services into forest management and planning are discussed.

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1. Introduction

The challenging nature of decision making and the number of decision alternatives in forest planning vary considerably depending on the planning problem at hand. For example, spatial scale may vary from the level of a tree to the level of the whole country and even to the whole continent. The time scale of planning can range from an operative decision that will be implemented immediately to strategic-level planning, where the development of a forest area is predicted several hundreds of years into the future (e.g. Kangas et al., 2015). Numeric and detailed inventory data form the basis for operational and tactical planning, whereas aggregated data dominate in strategic planning. In addition, qualitative data have a remarkable role in normative (political) planning when, e.g. the needs of the society are considered.

Forest management decision problems are often simplified by applying traditional growth and yield tables or general forest management recommendations (e.g. Äijälä et al., 2014). In this approach, the decision maker (DM) utilizes directly the forest inventory data supplemented with some variables derived from

these data such as the total timber volume or saw log volume of the stand. In addition to the simple approach, empirical models and simulation and planning systems have been developed to predict the outcomes and effects of future forest management alternatives (e.g. Borges et al., 2014). In modern planning systems, optimization methods are also utilized to construct alternative forest plans for a planning area.

A challenge in forest management planning is that several, often conflicting criteria, need to be taken into account when making decisions. Some decision problems are exploratory by their nature, i.e. instead of directly implementing the results, they are utilized in learning and providing understanding of forest dynamics for a given decision situation. Hence, the DMs' learning from the data and planning calculations needs to be properly supported before a well-argued decision can be made. The data must be managed so that meaningful comparisons and searches among decision alternatives are enabled.

Decision and planning problems can also be approached using data visualization. "Interactive visualization" refers to an information visualization process that supports the formation of mental image data, and this formation is further augmented (Ware, 2004) by allowing the user to interact with the data (Siirtola, 2007). For example, bar charts, scatterplots, multidimensional scaling, parallel coordinates and glyphs are typical techniques used for interactive visualization (e.g. Miettinen, 1999; Siirtola, 2007). Interactive data

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visualization is an increasingly popular approach for improving the interpretability of multidimensional data and supporting decision making in science and business (e.g. Oghbaie et al., 2016).

In operational forest planning, interactive visualization may support decision making for example in searching for appropriate forest stands from which energy wood or high-quality logs can be procured. In tactical and strategic forest planning, characteristics of alternative forest plans can be illustrated using interactive visualization, which helps also in the selection among the alternatives. The need to use interactive visualization in forest planning is also connected to the use of large datasets in calculations. These data could be examined in more detail and more efficiently by using visualization techniques that are not yet commonly used.

The ongoing bioeconomy development emphasizes the use of renewable natural resources such as wood, to produce energy and material (Spatial Foresight et al., 2017). Simultaneously, multi-objective land use is increasingly seen as an option to concurrently produce ecological, social, and economic benefits (de Groot, 2006). In the forest planning context, these ongoing, partially divergent trends emphasize the need to incorporate new criteria into calculations and visualizations. For example, the synergies and trade-offs of biodiversity, carbon balance and recreational services in relation to timber production can be illustrated to the DM. This helps in the forest management decision-making process, during which different forest management operations are targeted to the areas where they are best suited.

There are a few applications generated for multi-objective forest planning purposes, which utilize interactive visualization tools. For example, Pasanen et al. (2005) introduced an application of parallel coordinates for tactical planning of non-industrial private forests, and the same web-based tool was also used in strategic-level participatory planning of the use of natural resources in northern Finland (Hiltunen et al., 2009). Chertov et al. (2002) suggested the integration of forest simulation models with spatial-temporal interactive visualization into interactive maps that aid in landscape-level decisions. Rammer et al. (2013) introduced an adaptive web-based forest management tool, which included a graphical user interface in which a shifting between several graphical variants to explore climatic factors and management alternatives on the performance of ecosystem services was enabled. However, to our knowledge, there have not been any extensive surveys of the visualization techniques and applications that could be used rather straightforwardly in the forest planning context.

The aim of this paper is to introduce and evaluate the potential of existing interactive data visualization techniques to support decision making in increasingly challenging multi-objective forest planning situations where more outputs are demanded from forest ecosystems. To do this, we first present a forest planning framework related to the data characteristics corresponding to the planning situation. Second, we demonstrate 14 potentially suitable data visualization techniques using forest planning test data from example stands, and evaluate their strengths and weaknesses. When selecting the potentially suitable techniques, the aim was to select a diverse group of techniques potentially suitable for the forest planning framework presented below. Finally, we discuss the broader usability of interactive visualization techniques in the context of the renewable natural resources management planning.

2. Planning framework using interactive visualization

The planning framework applied in this study includes three levels, which are connected to different data that are typically available at these levels (Fig. 1). At the first planning level, decisions are made at the stand-level and they are based on the current state of the forest stand. Thus, available forest resource data are used at

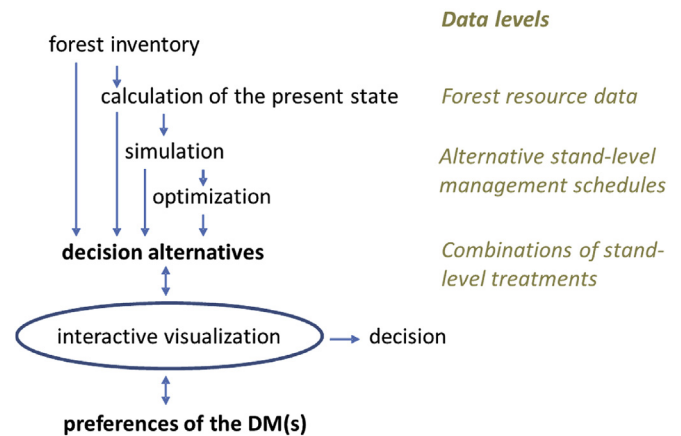


Fig. 1. The three-level forest planning data framework using interactive visualization of data.

this level. At the second level, decisions are still made at the stand-level, but there are possibilities to use more elaborated data, such as simulated treatment schedules for forest stands. At the third level, decisions are made based on holding-level data. Often, the first level serves the needs of operational planning, the second level, supported with planning system that includes optimization technique, serves often the needs of tactical planning. Finally, the third level that can include comparisons of holding-level alternatives is more strategic by nature. At all three levels, the planning process follows the steps of the multi-objective planning process defined by Kangas et al. (2015):

- Defining the decision situation and the decision problem. For example, the planning area and the DM(s) are defined during this phase.
- Clarifying the preferences of DM(s) regarding the use of forests. This planning step provides forest management goal variables and the corresponding goals to be strived for by applying interactive visualization techniques.
- Defining the decision alternatives and their outcomes regarding the goals of the DM(s). Depending on the planning level and the availability of data, outcomes may consist of (1) the calculated present state of forest stands based on inventory data, (2) outcomes of stand-level simulated treatments or (3) of alternative combinations for the stand-level treatments (Fig. 1).
- Evaluation of the decision alternatives by using interactive visualization.
- Choosing the best decision alternative.

All the above phases help the DM in the multidimensional task of defining how, when and where to manage the forests in different spatio-temporal planning situations. The DM can be a forest owner, a forest planner offering forest services for forest owners, or a wood procurement expert working for the forest industry.

The first data level, *forest resource data*, is often collected in field inventories. The data describe stand characteristics such as the stand area and boundaries, site fertility, and the main characteristics of the growing stock such as the tree species, their mean diameters, basal area, age, etc. These data are often further processed and updated to the present state so that, e.g., the inventory that was carried out in 2013 is updated to the year 2016 by predicting the growth of trees (e.g. Hynynen et al., 2002). In addition, several other variables can be calculated at this phase. For example, total volume and saw log volumes by tree species, growth and cutting value can

supplement the basic inventory data. The amount of data at this level depends on the size of the planning area: in privately owned forests where the typical forest holding is approximately 30 ha (Hänninen et al., 2011), the number of forest stands can be 20–40. The number of variables describing the present state depends on the planning situation but it can easily increase up to 50 different variables. In industrially owned forests, the amount of stands is evidently much larger. The use of the stand-level data still seems to have a dominating position in forest planning even though novel forest inventory methods such as laser scanning, digital photos and historical data allow versatile and cost-effective data processing that would support more advanced forest management decision making (Lehtomäki et al., 2015).

For the second data level, *alternative stand-level management schedules* for each stand are generated by utilizing the forest resource data and simulation models and tools (e.g. Bettinger et al., 2009; Kangas et al., 2015). Depending on the length of the planning period, stand characteristics and simulation instructions regarding the rules to carry out clear cutting, thinning, over-story removal, and other (pre)commercial forest management operations, a varying number of alternative management schedules are created for each stand at this level. The number of stand-level management schedules can vary from one to several hundred depending on the simulation instructions and the length of the planning period. For example, in the study by Kärkkäinen et al. (2014), the average number of schedules (including energy wood harvesting alternatives) for each management unit was 210 for a planning period that included four 10-year sub-periods. In addition to the outcomes of each management schedule, the resulting data also include the predictions of the state of the forest stand at the end of each sub-period. This means that the amount of data increases notably at this planning phase. However, at the individual stand level, despite the amount of stands, the forest owner's aim is to select one management schedule for each stand.

At the third level, the data are aggregated and one or several combinations of stand-level management schedules, i.e. discrete forest plans, are created. These alternatives are often generated by using optimization methods; most often by using linear programming (e.g. Weintraub et al., 2000), goal programming (e.g. Aldea et al., 2014) or heuristic algorithms (e.g. Bettinger and Zhu, 2006; Kangas et al., 2015). In comparisons between these alternatives, the decision criteria from phase (ii) of the multi-objective planning process are important. When the final decision is made, one management schedule has been selected for each forest stand of the planning area.

3. Demonstration of interactive visualization techniques in forest planning

In this section we demonstrate techniques that can be applied to visualize data available at the three data levels described above. Under each level, we first describe the main characteristics of the test data used in the example planning situation. Thereafter the visualization techniques potentially suitable for the planning level in question are demonstrated.

We scanned the existing visualization techniques from the literature and existing visualization software. As there are a wide range of visualization techniques, we chose the most potential ones for further evaluation. The selected techniques had to be suitable for the needs of multi-objective forest planning and for the data levels introduced above. In addition, they had to be able to accommodate efficiently several decision criteria and alternatives, and their usability and assimilation should be easy for decision makers and forest planners. Finally, we ended up evaluating 14 different techniques.

3.1. Data level 1: forest resource data

3.1.1. Test data and example planning situation

The forest resource test data for the case forest holding was based on airborne laser scanning (ALS)-based inventory with supporting field inventory. The case holding of 117.2 ha consisted of 126 stands in eastern Finland. The total volume of growing stock was 16 019 m³.

In our example planning case, the hypothetical owner of the case forest holding is interested in stands containing birch saw logs. In addition, the owner does not like large clearcutting areas. Thus, the owner is interested in stands where the amount and percentage of birch saw logs is high and also where the stand size is small.

In a typical planning situation and in printed forest plans, the planning data are visualized with tables, scatterplots or bar charts. Tables are familiar and popular for investigating and getting acquainted with planning data, and scatterplot is one of the most frequently used multidimensional data visualization techniques (e.g. Zheng et al., 2015). Besides tables and scatterplots, bar charts (histograms) are widely used visualization techniques.

3.1.2. Visualization 1: pairs plot

A multidimensional data set can be expressed as a pairs plot to offer a general view of the data (Fig. 2). The names of the (goal) variables are on the diagonal and pairwise scatterplots are projected over it. Different stands can be identified, for example, by labelling them. When paired plots are used interactively, the user can identify and also select those stands, where the stand properties best meet the search criteria. In principle, it is also possible to summarize the selections to support the DM if the aim is to find, for example, a certain amount of birch saw logs from the forest holding.

The problem with scatterplots, as well as other pairwise representations, is that they do not allow to visualize higher multidimensional relationships. Attempts to resolve the problem have been done through interaction and presentational methods like multidimensional displays with a 2D or a 3D appearance (e.g. Ahlberg and Schneiderman, 1994; Cui et al., 2006; Wilkinson et al., 2006).

3.1.3. Iconic techniques

With iconic techniques, the data are presented as images or glyphs in which some visual features are describing variables. This approach emphasizes a visualization of larger amounts of goal variables multi-dimensionally. Images can be abstract like star shapes (Mañas, 1982; Sobol and Klein, 1989), spider-webs (Kasanen et al., 1991) or more symbolic, such as familiar human faces (Chernoff, 1973).

Visualization 2, the star approach, expresses each circle as one alternative (stand), and the area of each star within circle indicates the goodness of that alternative (Fig. 3). In our test case, stand area differs from other criteria, as the smaller the stand area, the better the alternative stand is. Thus, larger stands are located near the origin. In a situation where all criteria are maximized, the star approach enables the rapid finding of stands, in which the area of the star is largest, i.e. in which the important decision criteria are performing best. To help this, a reference star based on the most desirable values of the decision criteria can be utilized to help the decision maker (Fig. 3). Thus, the size and the shape of each star directly show the overall goodness and relations of decision criteria for each stand.

Visualization 3, Chernoff's faces, are among the first approaches to visualize multidimensional data (Chernoff, 1973). Characteristics of the face, eye sizes, position of pupils, length of nose, etc., can represent up to 18 criteria values for each alternative. In Fig. 4, the

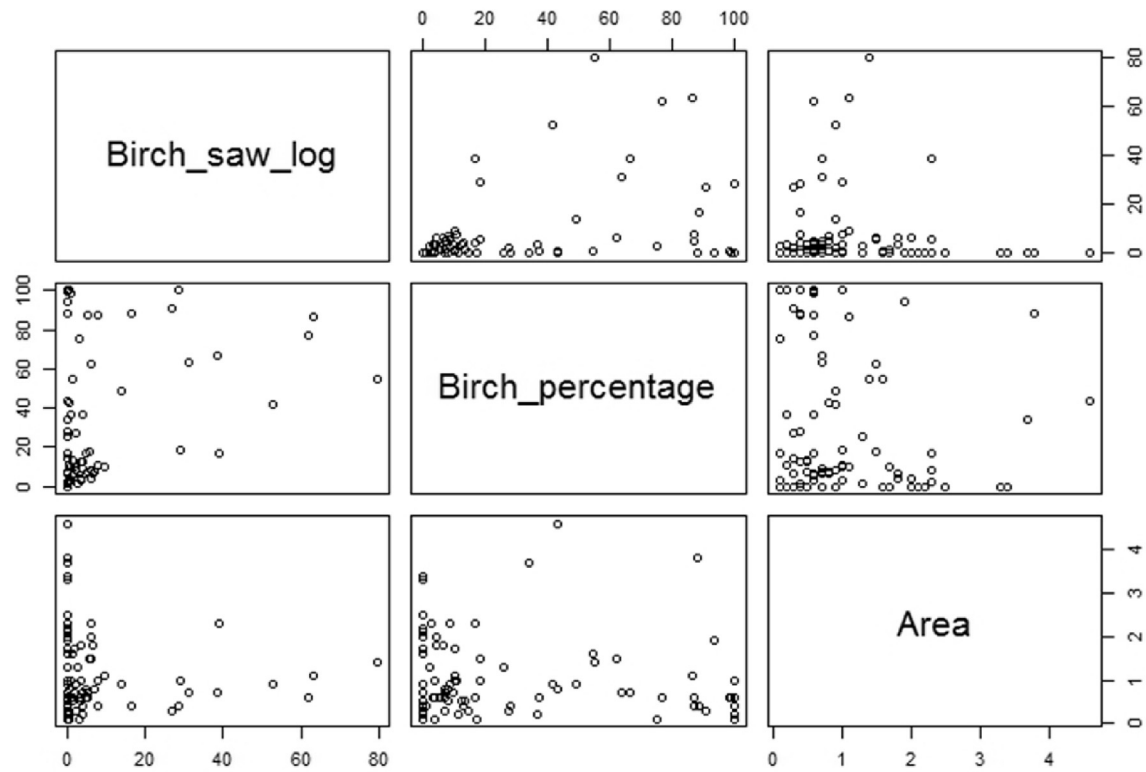


Fig. 2. Three goal variables from test data as a pairs plot. Each dot represents a forest stand.

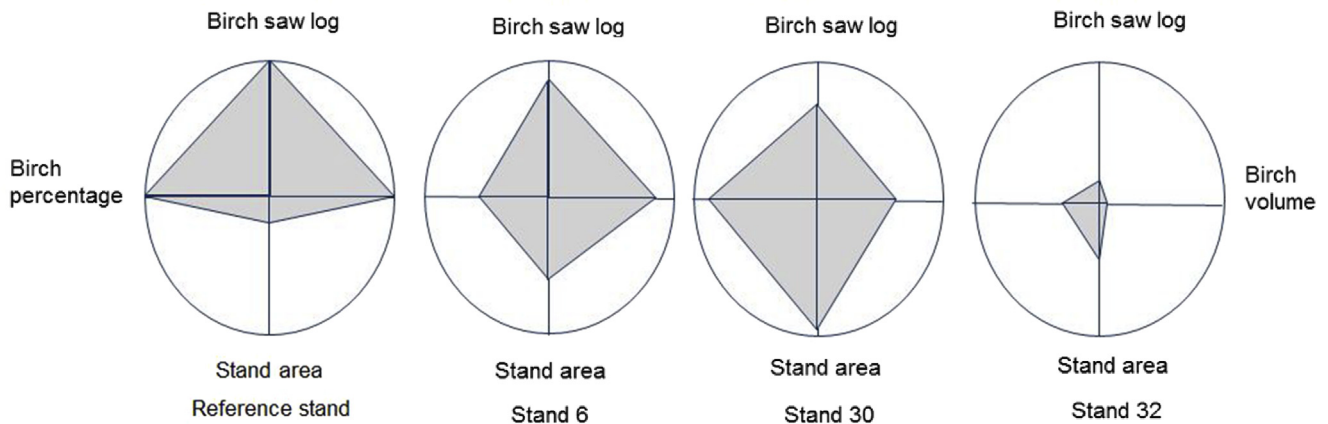


Fig. 3. Star icons of reference stand and three example stands.

length of the eyebrows is indicative of the criterion birch saw logs, and the size of eyes to birch volume, while stand area is expressed as the length of the nose (shorter the line, the better) and the percentage of birch as the length of the mouth. Chernoff's faces enable the DM to process these criteria in parallel and search for "certain kinds of faces" from the data based on his/her preferences.

There are a lot of other graphical objects or symbols than faces, like harmonious houses (Korhonen, 1991), metroglyphs (Anderson, 1960), stick-figures (Pickett and Grinstein, 1988) and arrows (Everitt, 1978), but the main principle is the same: their use facilitates the survey between the goal variable values and the visual features of the icons. The familiarity of used icons, like faces, makes interpretation easier (Jacob, 1978). Familiar icons can however also

introduce bias, because the DM may choose the management goals based on the attraction to certain types of faces. This can be avoided by letting DM first choose the 'ideal face', then ask his/her preferences about decision criteria, and finally adjust the decision criteria characteristics.

In interactive use, the DM can also arrange icons based on one goal variable or variables. The icons can also be arranged based on other classification criterion than goal variables, like forest holdings if a forest owner has more than one holding. In interactive glyph visualizations (e.g. Tufte, 1997), classifications based on other criteria are typically more effective than the ones depending on the glyph attributes. The drawback in the use of graphical objects presented above is that they are useful for a limited data (Siirtola,

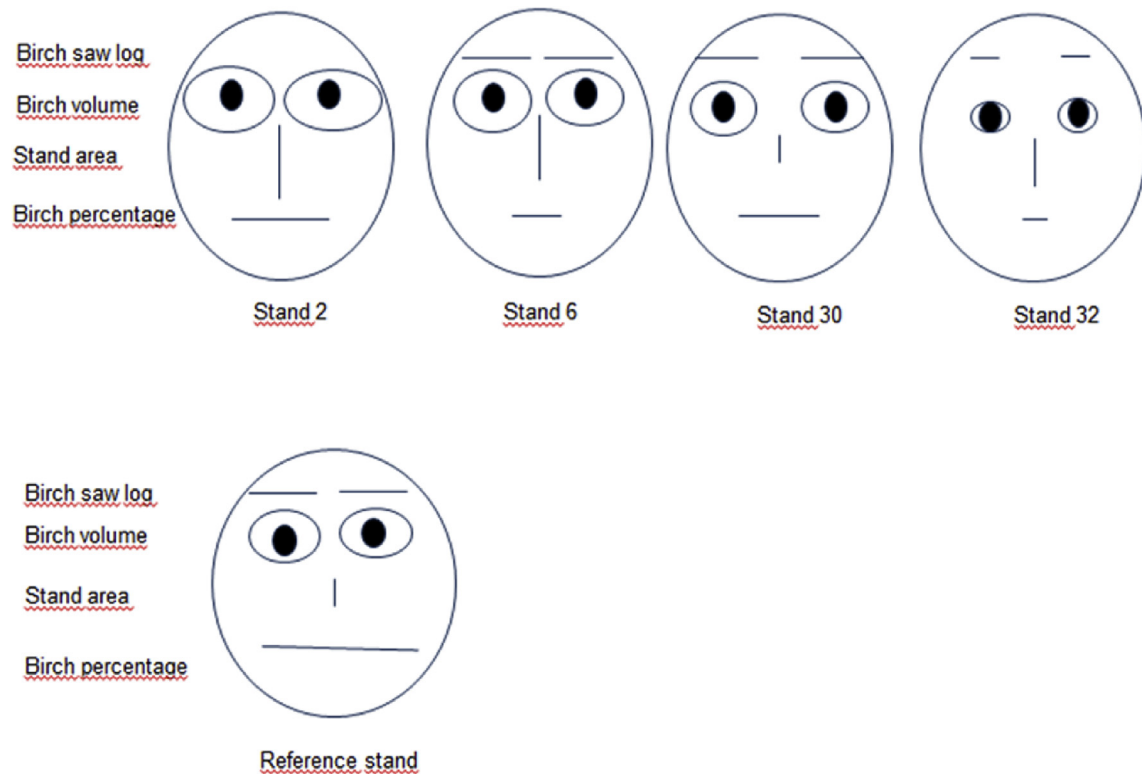


Fig. 4. Chernoff's faces of four example stands and one reference stand modified by forest owner based on his/her preferences. The length of the eyebrows expresses birch saw logs, the size of the eyes birch volume, the length of the nose stand area and the length of the mouth percentage of birch.

2007), i.e. the amount of stands under considerations should be quite small.

3.1.4. Visualization 4: data maps

Integration of forest simulation-optimization systems into interactive maps has been adapted to interactive forest planning (e.g. Pukkala, 2002; von Gadow and Pukkala, 2008). In forest planning, inventory data typically include the location of each stand, which can itself be an important decision criterion. In these situations, thematic maps with hues of colours, different textures, patterns, etc. have been traditionally used to illustrate geographical

division of certain characteristics. For example, in Fig. 5, the amount of birch saw logs is showed using the brown colour, the darker the stand, the larger the amount of saw logs. Birch percentages are also showed by the numbers inside stands, and the size of the stand and the distance from the nearest road can be seen directly from a map. If there are several criteria, adjacent maps can be drawn or data can be compounded onto one map using, for example, colours and textures together, or small bar charts or even tables within stands. If applicable, the use of different background maps, like topographic maps, can offer useful information for decision making. The utilization of maps helps DM(s) on focus selection of stands by the

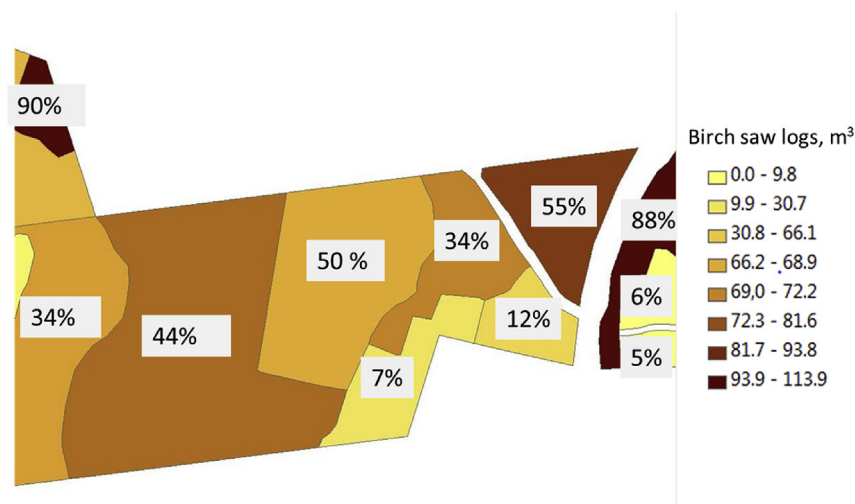


Fig. 5. Data map of the amount of birch saw logs in example stands. The darker the stand, the larger the amount of saw logs. The percentages of the birch are shown inside stands. The area of each stand and the distances from the nearest road (white line) can be estimated visually.

order of the volume of the wanted material or immaterial goods and locations of stands.

3.2. Data level 2: alternative stand-level management schedules

3.2.1. Test data and example planning situation

In the example forest holding, the alternative management schedules for stands were generated with a large-scale forest planning system MELA2009 (Redsven et al., 2011). MELA includes a forest simulator and an optimization package (Siitonen et al., 1996). The simulator automatically generates alternative treatment schedules for stands based on built-in event routines and forest management recommendations (Ruotsalainen, 2007; Äijälä et al., 2014). In MELA, the development phases of the growing stock (in-growth, growth and mortality of trees) are estimated using tree-level models (e.g. Hynynen et al., 2002). When calculating the economic outcomes from management schedules, stumpage values for different timber assortments and silvicultural and cutting costs were based on the average prices and costs used in Finland from year 2014 (Finnish Statistical Yearbook of Forestry, 2014).

To illustrate the use of visualization at the second data level, five alternative management schedules for a pine stand from the case holding were simulated for a 20-year period. Initial volume of the growing stock of the stand was 223 m³/ha with an annual growth 3.8 m³/a, and mean diameter of 24 cm. The simulated treatments were:

- 1) No management (REST)
- 2) Instant clearcutting (CLEAR)
- 3) Strip cutting 2015 (50% from the basal area), and clearcutting of the rest in 2035 (STRIP)
- 4) Small area cuttings (30% from the basal area), and clearcutting of the rest in 2035 (SMALL)
- 5) Thinning 2015, and clear cutting after 20 years (THIN)

Preferences of forest owner(s) define the visualization of goal variables. In this case, the selected criteria were the amount of deadwood in 2035, net incomes during the 20-year planning period, and discounted Net Present Value (NPV3) with a three percent interest rate (Table 1).

3.2.2. Visualization of a large amount of simulated management schedules

If the planning period is long and different forest management strategies exist that can be applied in stands, the number of simulated management schedules may become large. In this situation, the earlier-mentioned bar charts, scatterplots and glyph techniques can be used to discover differences between the outcomes of goal variables of various management schedules. They are familiar and simple techniques, but they also enable the illustration of the temporal development of stands depending on the applied management schedule. In addition, it is possible to use the above-presented pair plots in this phase. However, in interactive use, it must be noted that if the forest owner selects one management schedule from the pair plot, other management schedules from that stand must be excluded simultaneously. In addition to these techniques, also interactive maps (see Chapter 3.3) can be used to examine the stand-level management alternatives in a spatial context.

3.2.3. Visualization 5: decision tree

A decision tree is a decision support technique that uses a tree-like graph and model of decisions and their possible consequences, including, for example, chance event outcomes, resource costs, and utility. In Fig. 6, a simplified decision tree that illustrates the five

alternative management schedules from Table 1 is presented. The amount of deadwood in 2035 has been added to the outcomes. This technique combines graph visualizations and tables. The outcomes of the management schedule alternatives can be analyzed periodically. Decision trees are simple to understand and interpret, and allow the addition of new possible alternatives. They can also be combined with other decision techniques.

3.2.4. Visualization 6: multiway dot plots

In multiway dot plots, the bars of the bar chart of each alternative are arranged and located in the landscape orientation (Cleveland, 1994). Each alternative is presented like a panel, in which rows represent criteria. In Fig. 7, the multiway dot plot of the five alternative management schedules is presented so that the dots represent management schedule criterion values. When values between different criteria vary substantially, criteria values can be scaled. Furthermore, the values of the criteria of the alternatives can be shown, like the values of deadwood in Fig. 7.

3.2.5. Visualization 7: motion charts

A motion chart is an interactive and dynamic technique for visualizing large longitudinal multivariate data (Al-Aziz et al., 2010). Motion charts consist of common 2D Bubble charts for visualization of large multivariate data, and they further utilize supplement dimensions like the size and colour of the blobs (bubbles) and time for interactive visualization and demonstration of the data. The main object of a chart is a blob, which has three important attributes: size, position and appearance. Motion charts enable supervised temporal appearance of the blobs.

Motions charts enable visualization and interactive exploration of the relationships up to four goal variables of stand management schedules over a period of time in a simple but also an intuitive way. In Fig. 8, a motion chart of the five stand management schedules is presented from the years 2015 and 2031. Interactive characteristics of the chart are that the user of the chart can observe temporal development of decision criteria by moving the cursor on the time line. The colour of the circles indicates the amount of NPV3 of each management schedule from blue (low) to red (high). Furthermore, the user can decide on what way to illustrate decision criteria. In Fig. 8, e.g., the temporal development of the amount of deadwood has been expressed in two ways: by size of the circle and by the x-axis.

3.3. Data level 3: combinations of stand-level alternatives

3.3.1. Test data and example planning situation

Efficient, i.e. Pareto-optimal, forest plans are those whose decision criteria cannot be further improved without compromising at least one criterion. At the forest holding level, the amount of possible forest plans, i.e. combination of stand-level alternatives, becomes rapidly enormous. For example, if there were only two alternative management schedules in each stand, the total amount of possible holding level forest plans would have been $8.51 \cdot 10^{37}$ for the case study holding with 126 stands. In the example used here the simulation with MELA software produced 1501 alternative

Table 1
Goal variables of the five alternative stand-level management schedules.

Management schedule	Deadwood, m ³ /ha	Net incomes €/ha	NPV3, €/ha
REST	22	0	5600
CLEAR	8	7720	7720
STRIP	11	8580	6690
SMALL	13	9430	6270
THIN	20	8450	6760

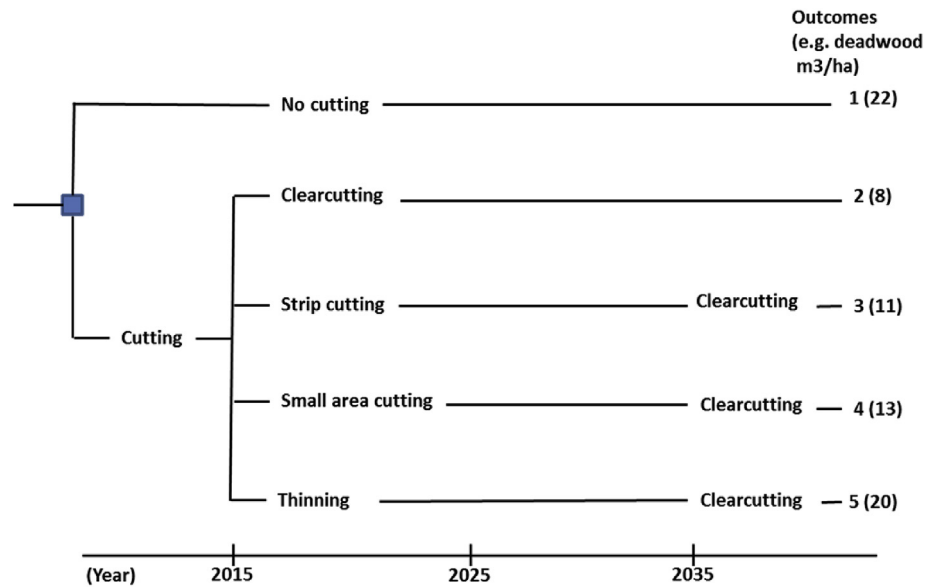


Fig. 6. Decision tree of the five stand-level alternative management schedules from Table 1. One example outcome, namely the amount of deadwood in 2035, is also shown in parenthesis.

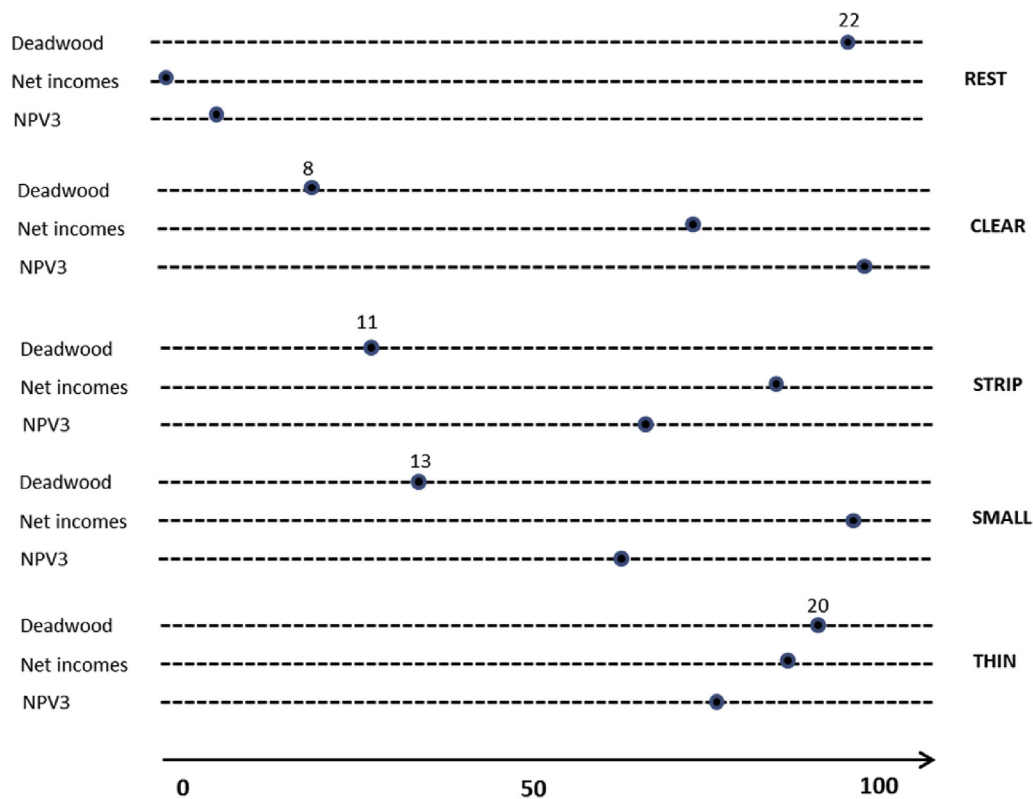


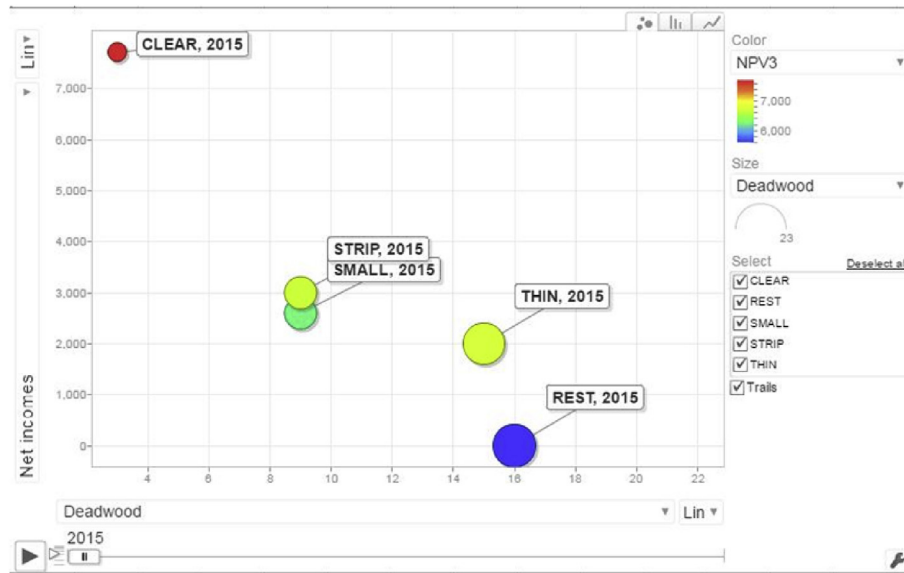
Fig. 7. Multiway dot plot of the five alternative stand-level management schedules.

management schedules, i.e. on the average 12 management schedules per stand. The creation of all efficient forest plans and their illustration would be infeasible using traditional optimization and visualization techniques for the example forest holding.

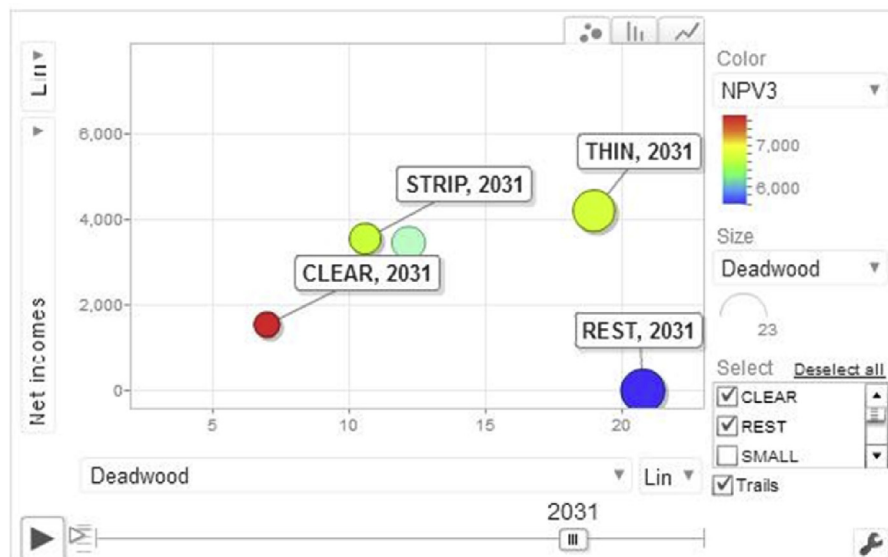
Therefore, in many forest planning processes, a forest planner creates one plan or a limited number of forest plan alternatives for further inspection. The idea is that the DM(s) can select the best

forest plan from a limited number of alternatives (e.g. Pykäläinen et al., 2007; Hiltunen et al., 2009) for the implementation or for further fine tuning (e.g. Haara et al., 2014).

For the case study holding, five alternative forest plans were prepared using the simulated alternative management schedules regarding the forthcoming 20-year planning period. LP optimization was used in this task, with following objectives and



a)



b)

Fig. 8. Motion chart of the five stand-level management schedules at the beginning (a) and at the end of the simulation (b) (Google-motion chart, 2016).

constraints:

- 1) maximization of NPV, no clearcuttings (NO CLEAR)
- 2) maximization of NPV, no clearcuttings on first period (NO CLEAR1)
- 3) maximization of NPV, sustainable development (BAU)
- 4) maximization of NPV, 3% rate (NPV3)
- 5) maximization of net incomes on first period (INCOMES1)

The third plan, business-as-usual (BAU) was based on the forest management recommendations in which the focus is mainly on timber production-oriented use of forests. The outcomes of these plans are presented in Table 2 regarding the decision criteria that

Table 2
Outcomes of the five alternative forest plans.

	Plan				
	NO CLEAR	NO CLEAR 1	BAU	NPV3	INCOMES1
Volume 2035 (m ³)	19216	15050	18084	14620	11411
Clear cut 1 (ha)	0	0	9.8	17.3	22.4
Clear cut 2 (ha)	0	22.4	4	10.3	3.1
Clear cut total (ha)	0	22.4	13.8	27.6	25.5
NPV3 2015 (€)	643306	653031	656772	660309	615376
NPV3 2035 (€)	837864	668109	788127	661257	555747
Saw log 1 (m ³)	1525	1690	1877	2646	2568
Saw log 2 (m ³)	1507	3730	2018	2776	3730
Net incomes 1 (€)	124520	135111	140667	192147	211858
Net incomes 2 (€)	244022	409308	300616	405953	280597

were important to the forest owner. At this planning level, data and existing interdependencies and correlations between criteria need to be illustrated to the forest owner using the most suitable visualization techniques.

3.3.2. Visualization 8: parallel coordinates

Parallel coordinates is a popular multivariate visualization technique which enables plotting large high-dimensional data points on a plane (Inselberg and Dimsdale, 1990). The technique represents the N-dimensional data set C with coordinates $c_1, c_2, \dots, c_N$, by points on N parallel axes which are linked with polylines. In Fig. 9, six decision criteria and their values in five example forest plans are presented as a parallel coordinate plot using the EDEN visual analytics toolkit (Steed et al., 2012, 2013). The DM has already set upper limits of the area of clearcutting. Thus, currently only two plans are feasible (thicker lines).

The main advantage of the parallel coordinates technique is that the number of data dimensions is restricted to the horizontal width of the display unit. Another advantage is that the correlations between goal variables in the dataset can be spotted and observed. The adjacent goal variables provide the most obvious information about correlations between them. In order to analyze goal variables which are not adjacent, interactions and graphical indicators may be needed. To address this, many modifications to enhance the interaction and comprehension of parallel coordinates have been suggested, such as dynamic axis re-ordering, histogram display, details-on-demand capabilities, axis inversion (Hauser et al., 2002), dynamic interaction techniques (Siirtola, 2007) and informative line shading schemes (Johansson et al., 2003). Interactive examples of parallel coordinates can be found e.g. in Chang (2017).

The main problem in utilizing parallel coordinates with large data sets is that by displaying a large amount of data, also useful information is missed. For instance, the presentation of a numerous overlapping lines of decision alternatives prevents the observations of correlations in the data. Researchers have been working on improving this technique for better data investigation and improved user-friendliness and interaction against occlusion by adding data clustering (Yang et al., 2002) or statistical compression like standard deviations and means (Siirtola, 2000), brushing (Zudilova-Seinstra et al., 2009; Rundensteiner et al., 2015), enhancing the transparency of the polylines (Wegman and Luo, 1997), etc. In Fig. 10, 1000 alternative forest plans from sample forest holding are expressed as parallel coordinates. One cluster has

been generated from the data by selecting only those plans in which NPV (2015 at 3% rate) is high and the total clear cutting area has been restricted to about 2/5 of the permitted area. The means and standard deviations of the selected decision criteria of the cluster can be seen inside the restriction bars.

Value paths are a popular modification from parallel coordinates suggested to represent criterion vectors (e.g. Geoffrion et al., 1972; Schilling et al., 1983). In this approach, the ranges of the criteria in the Pareto optimal set are also displayed. When the ranges are known, supplementary knowledge about the potential and the restrictions of the criteria are given in a quite easily understandable way.

3.3.3. Visualization 9: correlation visualization

The EDEN toolkit also includes a correlation analysis tool enabling a rapid visual judgement of the magnitude of relationships between variables (Steed et al., 2012). When the user interacts with the system, the correlation data are kept up-to-date (Fig. 11). The correlation boxes are coloured from red for positive and blue for negative correlations. The stronger the correlation is, the more saturated and distinctive the colour is. This may particularly help with large data sets.

3.3.4. Visualization 10: knowCube

The knowCube is a novel multi-criteria decision support system, which has the knowledge navigation component that contributes graphical database retrieval and interactive, goal-oriented problem solving (Trinkaas and Hanne, 2005). The graphical interface is a combination of spider web (e.g. Kananen et al., 1991), Pareto-optimal ranges and circular parallel coordinates (Fig. 12). It represents several axes in a star-shaped form, each axis representing, in our case, one decision criterion, e.g. t_p stands for clearcut area in first period. One forest plan described by multiple criteria can be represented in the figure as a polygon, resulting from connecting lines between the criterion values on neighbouring axes. Furthermore, it is possible to interactively navigate through the data of forest plans. Clusters can be constructed within the database interactively by delineating minimum and maximum borders for each decision range on the screen (Trinkaas and Hanne, 2005). Then all forest plan alternatives locating within define a cluster, like a light grey area (decision corridor) in Fig. 12. More examples of knowCube can be found in Trinkaas (2010).

In Figure, the currently visualized alternative is acceptable as it

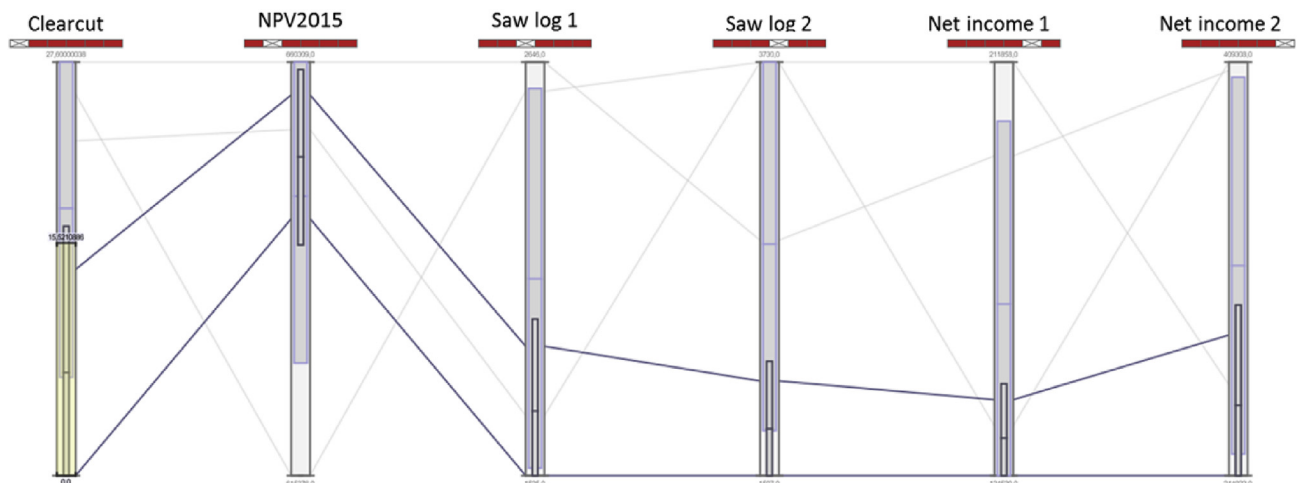


Fig. 9. The five alternative forest plans as a parallel coordinate plot (EDEN visual analytics toolkit (Steed et al., 2012)).

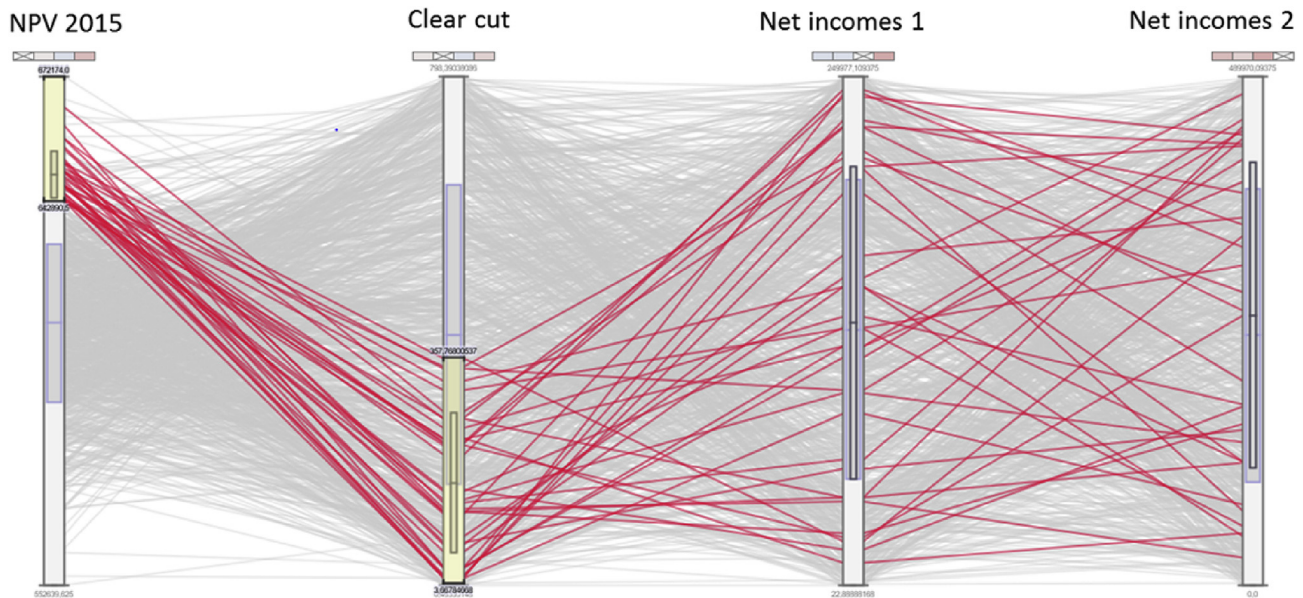


Fig. 10. Parallel coordinate plot of 1000 alternative forest plans from forest holding (EDEN visual analytics toolkit (Steed et al., 2012)).

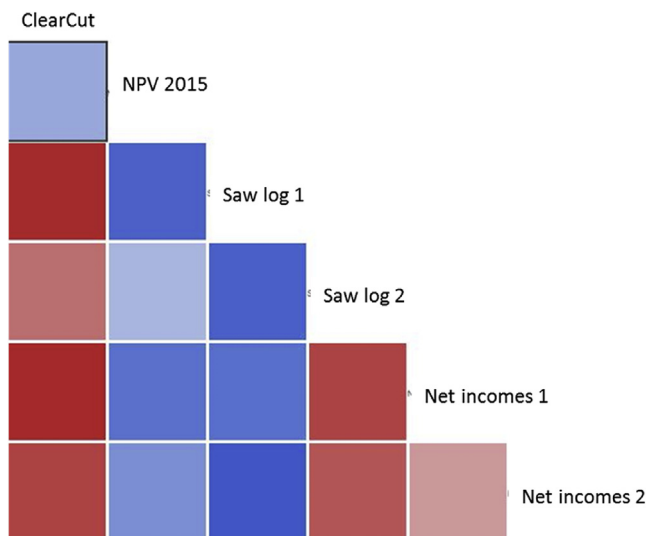


Fig. 11. Image illustrating correlations between six decision criteria of the forest plan data (EDEN visual analytics toolkit (Steed et al., 2012)).

is located inside decision corridor (light grey area). The alternatives are introduced one by one in the navigator, and the alternatives the DM finds potential can be saved for further examination. Criterion values are as follows: t_p clear cut area at first period, p_r volume at 2035, a_m clear cut area at second period, t_s net incomes at first period, t_b net incomes at second period, r_p NPV at 2015, r_2 saw log at first period and c_u saw log at second period.

3.3.5. Visualization 11. Tabular visualization techniques: reorderable matrix

In tabular visualization techniques, a powerful and universally known metaphor table is turned into an interactive image. The reorderable matrix method introduced by Bertin (1981, 1983) can be considered as the starting point of these techniques. The principle is simple: Multivariate data are presented more or less like a table. The data values are expressed with symbols, typically circles

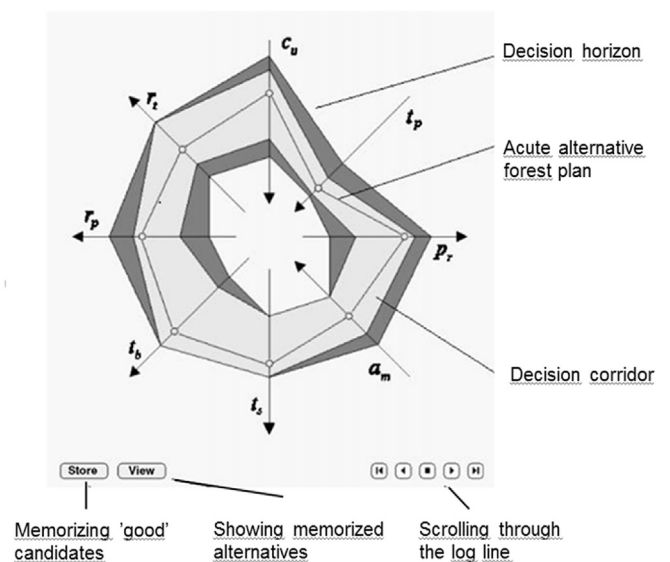


Fig. 12. KnowCube navigator (Trinkaas and Hanne, 2005).

or rectangles, which have a relative size depending on the actual variation of that variable.

Fig. 13 shows a reorderable matrix of three goal variables of the five forest plans. Basic technique can obviously not be applied to very large problems which have dozens or hundreds of alternatives. However, some standard graphical user interface techniques, such as panning or zooming, may help by providing simultaneous overview and detail views. Furthermore, different arrangement methods of the reorderable matrix data have been introduced interpreting the emerging patterns (e.g. Jain et al., 1999; Siirtola, 1999; Bertin, 2000; Mäkinen and Siirtola, 2000; Niermann, 2005).

3.3.6. Visualization 12: heatmaps

Interactive visualization techniques similar to the reorderable matrix method, namely heatmaps, were derived from techniques used in data mining (Lotov and Miettinen, 2008). Heatmaps are

Alternative	Volume 2035	Clearcut	NPV2015
NO CLEAR			
NO CLEAR1			
BAU			
NPV3			
INCOMES1			

Fig. 13. The reorderable matrix of the five forest plan alternatives (note that the goal variable clearcut is minimized, i.e. smaller values, indicated by larger rectangles are better).

matrices where each cell is coloured according to its value. The use of heatmaps offers a better information intensity than tables, and furthermore, it is easier to detect correlations and trade-offs between the criteria. In Fig. 14, each column in the matrix stands for one decision criterion and each column alternative plan. The easily understandable traffic light colour scheme from red low (high) values in criteria to be maximized (minimized) via yellow (medium values) to green (good values), was applied in this example.

3.3.7. Visualization 13: Pareto frontiers

Because typically the used objectives in complex ecosystem management problems are in trade-off between each other, it is practically impossible that all the objective values could reach their maximums at the same time. Consequently, the results of the optimizations can be presented with the production possibility boundaries, i.e. Pareto frontiers, in which none of the objective variable values can be improved without deterioration of at least one of the other objective variable value (see e.g. Borges et al., 2013; Kärkkäinen et al., 2014). Visualization of Pareto frontier informs about trade-offs (and synergies) between the objectives. In Fig. 15, the Pareto frontier, i.e. two-dimensional trade-off curve of the amount of deadwood at the end of the planning period and net incomes during the whole planning period, is presented. Furthermore, the locations of four alternative forest plans located on the trade-off curve or inside the feasible region are presented in Fig. 15.

It is not difficult to visualize a Pareto frontier of two to three decision criteria, but if there are more criteria, the visualization task becomes challenging. For example, Ruotsalainen (2009) has examined the integration of multi-objective optimization with the three dimensional (3D) virtual reality environment in order to study the Pareto-optimal solutions and approximated Pareto-optimal fronts. The virtual reality tool made comparison of solutions and navigation between solutions easier, and further, helped to visualize and analyze obtained solutions and reduced solutions needed to be computed for learning and understanding trade-offs of the conflicting objectives.

Borges et al. (2014) presented an adaptive search approach in which the Pareto frontier was first approximated, and then the feasible goals method was applied to receive a solution accepted by stakeholders from interactive decision maps offering the Pareto

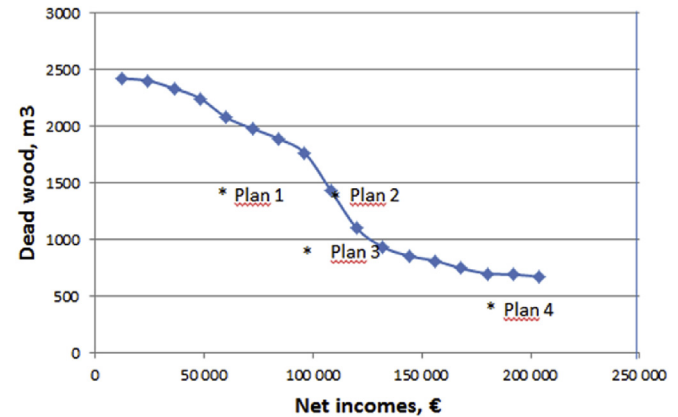


Fig. 15. Two-dimensional trade-off curve between the amount of deadwood at the end of planning period and net incomes in selection of forest plan. Forest plans 1, 3 and 4 locate inside the feasible region, while plan 2 locates on the trade-off curve.

frontier for forest management problems. The aims were to help the DMs assess trade-offs between objectives and set levels of achievement that reflect their preferences, and to generally demonstrate the potential of the techniques to provide a graphical representation of the feasible set in the criteria space when three or more criteria were considered.

3.3.8. Visualization 14: interactive maps

Interactions and trade-offs of decisions made for neighbouring stands are important, thus forest operations made for one stand may have an impact on operations of neighbouring stands, and in that way influence the outcomes of treatments of neighbouring stands in consequence of, e.g. scale-effects. Therefore, the locations of the stands, in which forest operations are performed, are also important.

The visualization of outcomes of alternative management plans can be performed on a sequence of interactive maps (Chertov et al., 2002). The maps enable direct representation of spatial patterns of forest dynamics under various alternative management plans. Fig. 16a and b shows the impact of two forest plans, BAU and NPV3, on the amount of deadwood. The difference between the two forest plans is visualized in Fig. 16c.

The approach enables many characteristics to be displayed on the map such as those related to biodiversity, e.g. species richness, amount of deadwood, and ecosystem services, e.g. carbon balance, indices of recreational attractiveness, and berry yields. The approach helps to analyze and understand the impacts of different alternative management plans at the landscape level. However, the visual analysis necessitates silvicultural, ecological and social skills (Chertov et al., 2002).

The adaptive forest management (AFM) ToolBox (Rammer et al., 2013) also applies interactive maps. The results of optimizations can be displayed and explored as a summary table, or in a more detailed view, but furthermore, the software enables visualization

Alternative	Volume2035	Clearcut	NPV2015	NPV2035	Sawlog1	Sawlog2	Netincomes1	Netincomes2
NO CLEAR	19216	0	643306	837864	152.5	150.7	124520	244022
NO CLEAR1	15050	22.4	653031	668109	169	373	135111	409308
BAU	18084	13.8	656772	788127	187.7	201.8	140667	300616
NPV3	14620	27.6	660309	661257	264.6	277.6	192147	405953
INCOMES1	11411	25.5	615376	555747	256.8	373	211858	280597

Fig. 14. Heatmap of the five forest plans with respect to decision criteria.

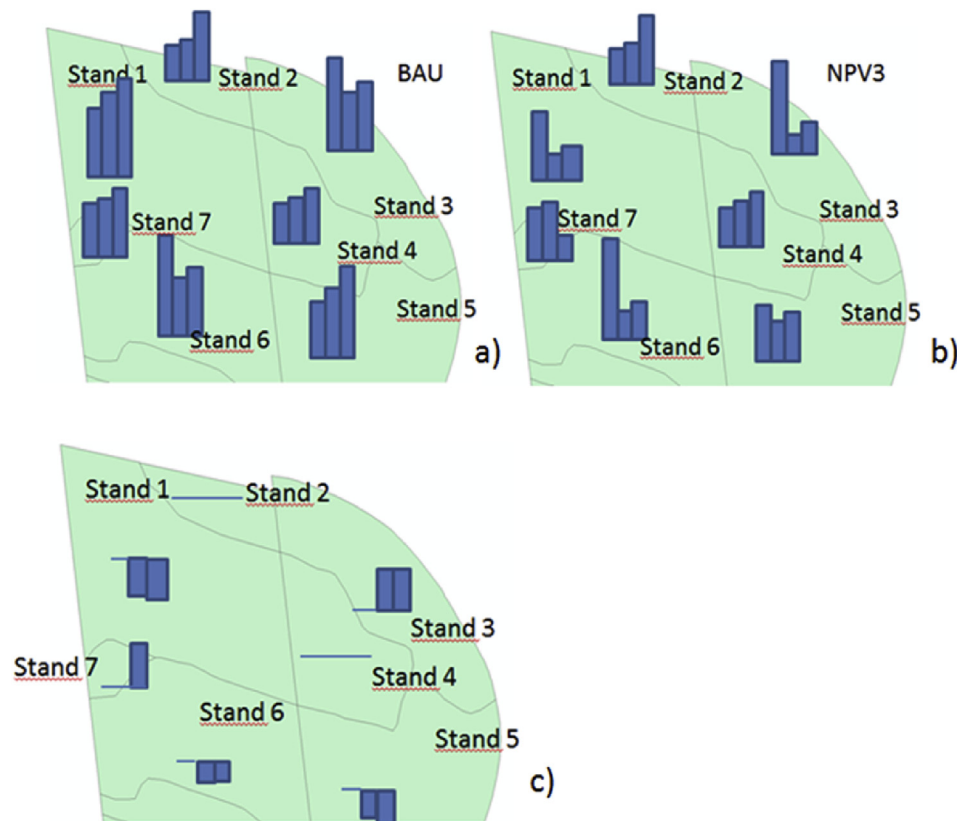


Fig. 16. Visualization of simulated time series of deadwood with forest plan alternatives BAU (a) and NPV3 (b) and differences between forest plans BAU and NPV3 about deadwood (c).

of optimized management plans at the level of management units or landscape on top of a displayed Google Maps platform.

Forest visualization can be used to improve the understanding and communication of the spatio-temporal impacts of the management alternatives as a part of interactive planning (Tyrväinen and Uusitalo, 2005). If visualization is combined with interactive optimization, forest owners can instantly see the effects of alternative management plans on the screen. This kind of service can also be offered over the Internet. Visualization enables the forest owner to compare the financial benefits of each area of interest, and helps the owner to better separate monetary benefits with non-monetary ones (Orland et al., 2001).

4. Discussion and conclusions

Forest owners are encouraged to participate in the planning process in order to define management proposals that match their goals and needs. In principle, optimization methods offer a technically effective tool for forest planning and they also serve in the production of solutions for owners. Interactive data visualization may facilitate this process in several ways. For example, visualization helps in profound and holistic understanding of the decision space, which, in turn, helps to formulate the optimization problem. Holistic evaluation supported by interactive visualization also gives the user a direct link to the data and information, and by doing this, improves the transparency and controllability of the selection process and helps the forest owner to be able to select the most preferred alternative.

Various different data visualization techniques are available for the needs of multi-objective forest planning. We present here 14

techniques, but there are dozens of other techniques such as projection-based approaches (Miettinen, 2014) principal components (Mareschal and Brans, 1988) and biplot (Everitt, 1978; Gabriel, 1981). New techniques are also developed frequently. The visualization techniques presented in this paper are summarized from the needs and properties of data levels and different forest planning situations in Table 3.

As Table 3 reveals, the introduced visualization techniques are suited to different forest planning situations depending on the amount of data and the level of planning. None of them is superior to others, but some are more suitable for certain decision problems than the others. However, the planner/expert always has to make a preliminary choice of which technique to apply. One of the presented techniques, namely parallel coordinates, is suitable for visualization of moderately large data amounts. However, it is possible, for example, to sort stands or cluster them first depending on the criteria, and then visualize only the most interesting ones.

Because of clearly different units and scales of decision criteria used particularly in forest planning, the use of some visualization techniques can be demanding. Changing a measurement unit to another unit in one criterion to better fit it to others may not be helpful if the use of unit has become established, like hectares (or acres). In these cases, for example parallel coordinates or icons are more illustrative than bar charts. However, instead of using absolute scales, it is also possible to use relative scales (Miettinen, 2006), as has been done with glyph techniques.

Visualization tools that can be used independently or aggregated to forest planning systems should be user friendly and not too complicated. Highly advanced and sophisticated techniques and tools are not helpful if they cannot be easily applied by a forest

Table 3

Summary of the introduced 14 visualization techniques and their evaluation against data levels. Abbreviations are explained after the table.

	Data Level	Max number of criteria	Max number of alternatives
1. Pairs plot	1 3	~5	100–500
2. Star icons	1 2 3	~10	n
3. Chernoff's faces	1 2 3	17	n
4. Data map	1	~10	n
5. Decision tree	2 3	~10	~10
6. Multiway dot plot	2	~10	n
7. Motion charts	2 3	4 + 1	~10–20
8. Parallel coordinates	3	~10	n
9. Correlation visualization	3	~10	n
10. knowCube	3	~10	n
11. Reorderable matrix	3	~10	n
12. Heatmap	1 2 3	~10	n
13. Pareto frontiers	3	2 (~10)	n
14. Interactive maps	2 3	~10	~10

Level 1: Forest resource level.

Level 2: Alternative stand-level management schedules.

Level 3: Combinations of stand-level alternatives.

n: Innumerable.

manager (e.g. [Rammer et al., 2013](#)). Applied visualization techniques should focus on user-specific needs, thus these techniques should be easily understandable to DMs, and at the same time, their adaptiveness and interactivity should be as user friendly as possible without compromising feasibility. All technical details and access to the data should be hidden in the background, and any useless information should not be included. The familiarity of the used colours, symbols, etc. in the visualization helps DMs rapidly understand the differences between decision alternatives.

Modification of the presented visualization techniques based on actual planning situation is advisable, when it actually increases understanding of the decision problem and consequences of the decision alternatives. For example, application of Chernoff's faces, as well as other glyphs, from static visualization towards more interactive use can be done by letting a DM first define the ideal face, and then decision alternatives can be scaled based on that.

The visualization tools may also supplement each other, but it is worthwhile to give DMs a final opportunity to select those techniques (s)he finds most beneficial and understandable. Still, there should be some preliminary choices made by software designers or planners, so that the DMs are not overloaded with having to solve additional decision problems. In addition, there is a risk that interactive visualizations could actually enhance the human tendency to see causal relationships where there are not any. When this risk is present, planners must choose carefully the design of visualizations.

The problem with large data sets is that typically traditional multivariate visualization techniques lose their effectiveness when more than a few hundred or thousand data points are displayed ([Yang et al., 2002](#)). For example, the number of alternative treatment schedules and forest plans can be very high depending on this. There can be a voluminous number of feasible plans at the beginning of the decision process, which may cause a reduction of decision space by using non-compensatory strategies like e.g. lexicographic rules, elimination-by-aspect, or conjunctive strategies (e.g. [Kottemann and Davis, 1991](#); [Korhonen et al., 1997](#); [Chu and Spire, 2003](#)).

In most of our 14 example techniques, interactive operations zooming, sorting and clustering can be effective methods for handling big data sets. For example, in a parallel coordinate browser a set of selected lines can be represented with a single line showing the mean and deviation of the lines as proposed by [Siirtola \(2000\)](#). For example, [Fua et al. \(1999\)](#) applied visual hierarchies for occlusion reduction of the large data sets of the parallel

coordinates. [Yang et al. \(2002\)](#) suggested a general framework against clutter problem, in which a multi-resolution view of the large data set via hierarchical clustering was developed and hierarchical variations of four traditional multivariate techniques, namely scatter plot matrices, dimensional stacking, parallel coordinates and star glyphs stacking ([LeBlanc et al., 1995](#)), were used to show aggregation information about the resulting clusters. The presented method allowed filtering and navigation through desired focus regions at different level of details.

Because different visualization techniques enable seeing different things from data, hybrid approaches can improve interaction and gain proper insight into the data (e.g. [Siirtola, 2007](#)). For example, [Mosavi \(2013\)](#) suggested an interactive hybrid technique, which compounded finding of Pareto-optimal set and flexible visualization of the data set for example by parallel coordinates and multidimensional graphs. The technique was found to be capable of handling the large data sets and, with utilizing the features such as parallel filters and clustering tasks, it supported the DM interactively. [Siirtola \(2003\)](#) linked conceptually different graphics, i.e. parallel coordinates and reorderable matrix, together. Despite the linking initially slowed the process, it catalyzed learning and was well received by the users.

To conclude, it is not possible to rank the introduced techniques. However, some recommendations can be given from the results. For the first data level, data maps offer quick overlook to the forest resource data and extra information about location and surroundings of the stands. This may help especially in land management decision problems such as forest planning, when location can also be crucial in final decisions. For the second data level, motion charts offer time schedule that show interactively what are the incomes and the consequences of alternative stand-level management schedules. For the third data level, at least knowCube, Pareto frontier and parallel coordinates are potential techniques. knowCube and spider webs generally offer quick and illustrative technique to choose acceptable plans for further considerations. Whereas Pareto frontiers help decision makers to examine trade-offs of different, more or less controversial, decision criteria. When data are large, interactive use of parallel coordinates can help us to narrow the decision space. In addition, parallel coordinates offer an efficient way to compare high dimensional criteria together and to see the relationships between them. Of course, in the final selection of the intended visualization techniques, the preferences and skills of the DMs as well as the availability of the techniques have to be taken into account.

Regarding future prospects related to the use of visualization techniques, the need to incorporate ecosystem services into forest planning, management and decision making is growing along with the increasing awareness of the non-wood benefits of forests. Many challenges need to be solved associated with attempts to that direction. For example, a coherent and integrated approach to ecosystem service assessment and valuation is still lacking, and empirical data are still scarce (de Groot et al., 2010). Mapping of biodiversity and ecosystem services such as recreational value and timber production has been carried out over large spatial scales, such as at the European level (Maes et al., 2011, 2012). Forest planning however requires considerably more detailed assessment of ecosystem services and analysis of associated trade-offs; this process is still at an early stage. Unfortunately there is a tendency to reduce inventory work, which in turn, narrows possibilities to collect more versatile data.

To overcome the limitations considering for example detailed biodiversity data, habitat suitability modelling can be used to locate the environmental conditions suitable for key species from which there are no sufficient data (Kurttila et al., 2002; Aizpurua et al., 2015; Kangas et al., 2016). Recreational values can be mapped and quantified using public participatory GIS (Brown and Kyttä, 2014), and the integrative, still relative uncommon approach is to analyze the interactions and trade-offs of the mapped social-ecological values and create hotspots that require specific attention during planning (Alessa et al., 2008; Tolvanen et al., 2014). The next stage is probably to incorporate the hotspot approach into interactive maps that visualize the impacts of alternative management plans on timber production, biodiversity and ecosystem services. Thus, the techniques presented in this article provide approaches to visualize ecosystem services and their trade-offs so that sound decisions can be made regarding forests management.

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